

Unit-3

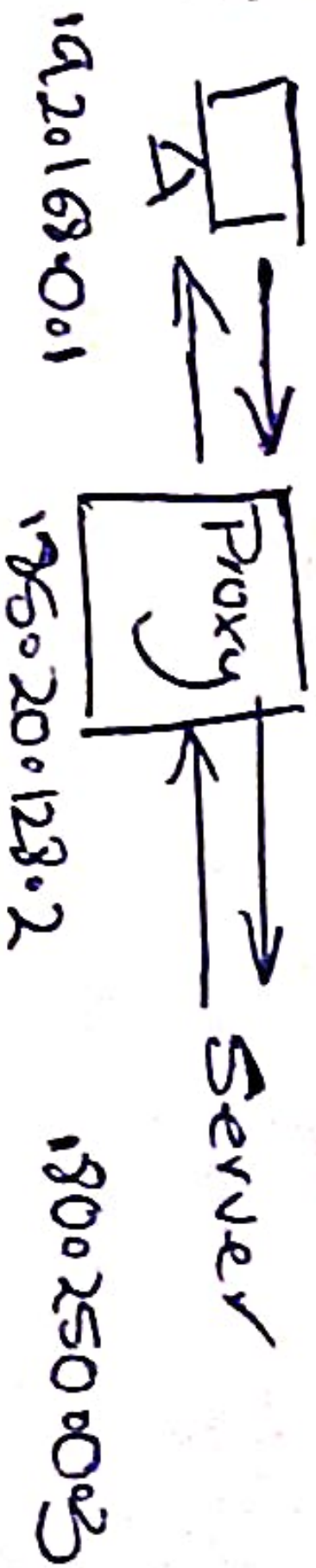
Online Privacy, Anonymity
& Secure browsing
concepts

Proxy Server :



A Proxy Server is a device or a service which acts as a intermediary between 2 devices.

-> Basic functionality is to hide your identity (IP address)



Proxy Server uses:

→ To hide IP Address
→ To ban the access of
certain web sites of
network administrator
ex capt.

→ By developer
→ Reducing traffic for trouble
making
→ Local Servers on Server
as proxy.

Proxy can be used
by:
~~the~~

General users

Network administrators

Developers

Proxy Disadvantages:

- Anonymity can be discussed
- Not encrypted (HTTP)
- keeps logs of all requests.
- Slow Speed

To eliminate Problems
of Proxy VPN Introduced

VPN ?

VPN is also work as
Proxy but

Encryption (Strong)

Speed (Med to Fast)

Coverage Full device

anonymity → High

over of sin.

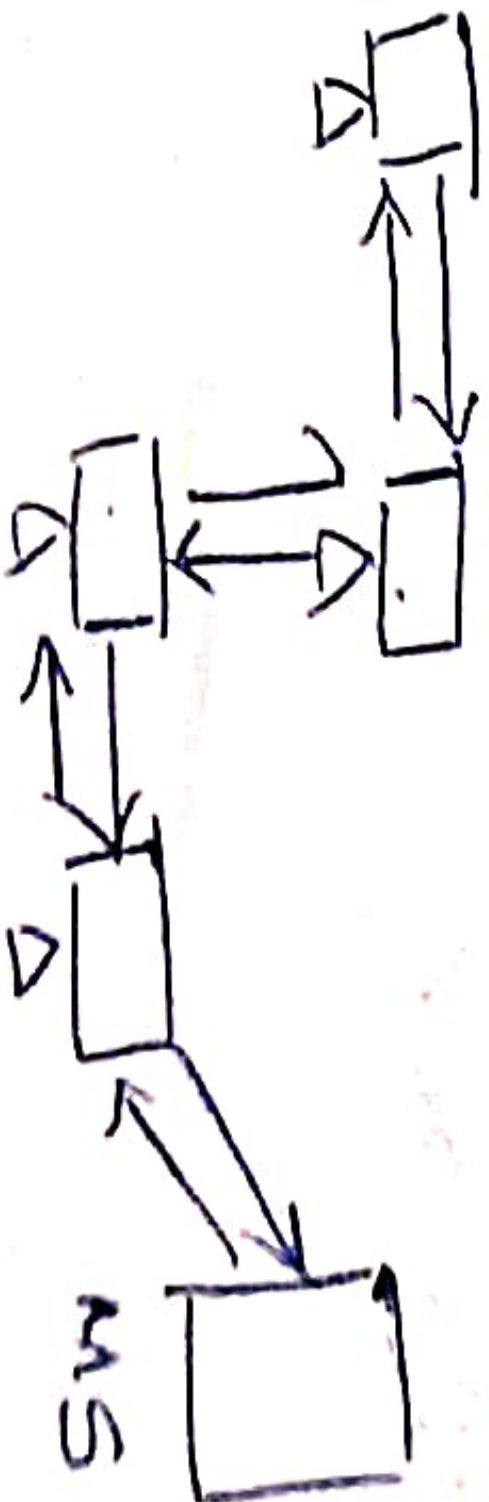
- Don't Save Logs
- Guarantee ~~any~~ anonymity
- Secure & Fast
- Fast connection
- To hide IP address from hackers, snoopers etc
- avoid region wide blocks
- avoid bandwidth restrictions
- mostly used by tor network

VPN Paid / Free
commonly used
VPNs

HotSpot Shield
Express VPN
Hide me

Working of Tor
Brouser

For uses decentralized
network of proxies



Proxy chain concept

Trace is difficult but
not impossible.

Why only 3 hops?

Note: • Onion sites uses
6 hops

3 hops for user anonymity.
6 hops website anonymity.

- ① Entry Guard
- ② Middle
- ③ Exit

Node	Knows your IP	Knows destination	Knows content
①	✓	X	X
②	X	X	X
③	X	✓	only if not hit

Comparison

VPN

Proxy

Tor



Feat	P	V	T
Encryption	NO	Yes (strong)	Yes (multiple layers)
Speed	Fast	medium to Fast	Very Slow
Coverage	only App / Browser	Full Device	only for Browser
Anonymity	Low	High	Extremely High